# Grade 10 – Test 2

**Name: _____**
**Grade / Class: 10**

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## English Language & Literature (15 Questions)

1. **Read the sentence: "Sara completed the assignment before the deadline."**
   **Question: What did Sara complete?**
2. **Identify the error in: "Neither of the boys have finished their homework."**
3. **Fill in the blank: "She ___ (study) French for three years before moving to Canada."**
4. **Rewrite in passive voice: "The chef prepared a special dish for the guests."**
5. **Choose the correct antonym: “Generous” – a) Stingy b) Kind c) Noble d) Brave**
6. **Write a dialogue between two friends discussing the importance of reading newspapers.**
7. **Read the passage: "Maya enjoys swimming and painting during her free time."**
   **Question: What hobbies does Maya have?**
8. **Identify the figure of speech: "The city never sleeps at night."**
9. **Write a descriptive paragraph about your favorite season.**
10. **Correct the punctuation in: "where are you going asked Ali"**
11. **Fill in the blank with the correct word: "He is interested ___ astronomy and physics."**
12. **Analyze: "The river sang as it flowed through the valley."**
    **Question: What literary device is used?**
13. **Convert to indirect speech: "I will join the meeting tomorrow," said Omar.**
14. **Choose the correct word: "There are ___ books on the table." (a) much b) many c) few d) little**
15. **Summarize the paragraph: "Forests provide oxygen, shelter for wildlife, and resources for humans. Protecting them ensures ecological balance."**

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## Mathematics (15 Questions)

16. **Solve: 3x2−5x−12=03x^2 - 5x - 12 = 03x2−5x−12=0**
17. **Simplify: (4x2y3)(3xy2)(4x^2y^3)(3xy^2)(4x2y3)(3xy2)**
18. **Find the slope of the line passing through points (-2, 3) and (4, -1).**
19. **Solve the inequality: 7x+2<3x+107x + 2 < 3x + 107x+2<3x+10**
20. **Find the area of a triangle with vertices A(0,0), B(4,0), C(2,3).**
21. **Evaluate: cos45∘+sin30∘\cos 45^\circ + \sin 30^\circcos45∘+sin30∘**

22. **Probability: A bag contains 3 red, 5 blue, and 2 green balls. Find the probability of picking a blue ball.**
23. **Find the volume of a cone with radius 5 cm and height 12 cm.**
24. **Solve for x: 5x−2x+3=1\frac{5}{x} - \frac{2}{x+3} = 1x5−x+32=1**
25. **Expand: (2x−3)3(2x - 3)^3(2x−3)3**
26. **A train travels 180 km in 4 hours. Find the speed.**
27. **Find mean, median, and mode: 8, 10, 12, 8, 15, 10, 8**
28. **Determine x: cot x=1\cot x = 1cotx=1 (x in degrees)**
29. **Equation of a line perpendicular to y = -3x + 7 through (2,1).**
30. **Solve: |4x+1|=9|4x + 1| = 9|4x+1|=9**

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## Physics (10 Questions)

31. **Explain why an object in free fall accelerates towards Earth.**
32. **State Newton's second law of motion with an example.**
33. **Calculate the work done when a force of 50 N moves a box 2 meters.**
34. **Define density and provide its SI unit.**
35. **A car increases speed from 10 m/s to 25 m/s in 5 seconds. Find acceleration.**
36. **Draw a ray diagram for the formation of a virtual image by a concave mirror.**
37. **Differentiate between mechanical and electromagnetic waves.**
38. **Explain the function of the cornea in human vision.**
39. **An electric bulb has resistance 12 Ω. If current is 2 A, calculate voltage.**
40. **Explain why rubber is an insulator.**

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## Chemistry (10 Questions)

41. **Define an element and give an example.**
42. **Balance: Al+O2→Al2O3Al + O_2 \rightarrow Al_2O_3Al+O2→Al2O3**
43. **Distinguish between acids and bases.**
44. **Explain the process of electrolysis.**
45. **Calculate the molar mass of H2SO4H_2SO_4H2SO4.**
46. **Identify the functional group in acetone.**
47. **Describe exothermic reaction with an example.**
48. **Name the gas produced when calcium reacts with water.**
49. **Define reduction with one example.**
50. **State Charles' law.**

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## Biology (10 Questions)

51. Explain the flow of blood in arteries and veins.
52. Describe the process of transpiration in plants.
53. Define habitat with two examples.
54. Name three parts of a neuron.
55. Compare meiosis and mitosis.
56. Identify xylem and phloem and their functions.
57. Briefly explain human digestive system.
58. Define conservation of biodiversity.
59. Name two viral diseases in humans.
60. Explain photosynthesis in one equation.

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## Social Studies / History / Geography (10 Questions)

61. Explain the causes of World War II.
62. Define human rights and give two examples.
63. Name three types of rocks and one characteristic of each.
64. Explain the role of UNESCO.
65. Identify three major ocean currents.
66. Describe the effects of industrialization on environment.
67. Explain the significance of the Enlightenment.
68. Define inflation and its impact.
69. Describe the Cold War rivalry between USA and USSR.
70. Differentiate between renewable and non-renewable energy sources.

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## Computer / ICT (10 Questions)

71. Define hardware and give two examples.
72. Differentiate between primary and secondary memory.
73. Explain flowchart and provide one example.
74. Define DNS and its purpose.
75. Name three output devices.
76. Explain email client vs. webmail.
77. Define malware and give one type.
78. Write Python code to check if a number is even.
79. Compare Wi-Fi and Bluetooth.
80. Define encryption and its use.

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## Islamiyat / Religious Studies (10 Questions)

81. List the Six Articles of Faith in Islam.
82. Explain the importance of Hajj in Muslim life.
83. Describe one moral teaching from the Quran.
84. Define Sadaqah and its significance.
85. Name two female Prophets or important figures in Islam.
86. Explain the meaning of Risalah.
87. Describe the purpose of fasting in the month of Muharram.
88. Name two major events in Islamic history.
89. Explain the concept of justice in Islam.
90. Discuss one way Islam promotes social welfare.

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## Extra / Mixed Knowledge (10 Questions)

91. Name three forms of energy transformation.
92. Define dictatorship and one disadvantage.
93. Name two endangered plants and reason for threat.
94. Solve: If 7 pens cost $21, how much do 12 pens cost?
95. Identify the continent with the largest area.
96. Explain the difference between tornado and hurricane.
97. Name two famous female scientists.
98. Write a short note on the importance of nutrition.
99. Convert 100°F to Celsius.
100. Discuss one major contribution of the Scientific Revolution.